

Test 3 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.DP.1.5 challenging	1	Count how many values fall inside that interval, including both endpoints.
2	MA.6.AR.2.2 challenging	$x - 19 = 34$	Starting amount minus the stickers given away equals what is left.
3	MA.6.AR.3.2 challenging	60 miles	A unit rate tells how much for exactly one unit, so divide miles by hours.
4	MA.6.GR.2.3 challenging	3	Divide the volume by the area of the base to find the height.
5	MA.6.DP.1.2 challenging	5	The median is the middle value when the data are in order.
6	MA.6.NSO.3.2 challenging	144	Add inside the parentheses first, then multiply by the factor outside.
7	MA.6.NSO.2.1 challenging	5	Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.
8	MA.6.AR.2.4 challenging	$1/5$	Divide the product by the known factor.
9	MA.6.AR.1.4 challenging	$6x + 12$	Only like terms can be combined, so add the x-terms and leave the constant.
10	MA.6.GR.2.4 challenging	180	Add the square base to the four triangular faces, each with area one half times base times slant height.
11	MA.6.DP.1.6 challenging	16	The new value becomes the greatest value, so the range grows.
12	MA.6.NSO.4.1 challenging	-7	Subtracting a number is the same as adding its opposite.
13	MA.6.AR.2.1 challenging	22	Only numbers greater than 19 make the inequality true, and 19 itself does not.
14	MA.6.NSO.3.1 challenging	180	The next time they leave together is the least common multiple of the two intervals.
15	MA.6.AR.3.5 challenging	0.5	Divide the price by the number of bottles to compare unit prices.
16	MA.6.AR.3.3 challenging	33	The total is the sum of the two parts in the same row.
17	MA.6.AR.1.1 moderate	$10x$	Match each word in the description to an operation in the expression.
18	MA.6.DP.1.3 challenging	8	The interquartile range is the upper quartile minus the lower quartile.
19	MA.6.NSO.4.2 challenging	-5	Divide the absolute values, then decide the sign from the two signs.

#	Benchmark	Answer	Explanation
20	MA.6.DP.1.4 challenging	8	The range is the greatest value minus the least value shown on the plot.
21	MA.6.NSO.2.3 challenging	\$69.35	Find the cost of all the items first, then subtract that from the amount paid.
22	MA.6.NSO.1.1 easy	-2.0, -0.25, 0.5	Each small tick mark is one fourth of a unit, so count fourths from 0 to place each number.
23	MA.6.AR.2.3 challenging	$11x = 99$	The number of baskets times the guavas in each basket is the total.
24	MA.6.NSO.2.2 challenging	$2 \frac{9}{20}$	Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.
25	MA.6.AR.3.1 challenging	5:11	A part-to-whole ratio compares one part to the total of all the parts.
26	MA.6.AR.1.2 moderate	44, 47, 48	Test each value in the inequality and keep the ones that make it true.
27	MA.6.GR.1.1 moderate	(-5, -5), (5, -5), (-5, 5)	Start at the origin, move along the x-axis first, then up or down.
28	MA.6.NSO.3.3 challenging	$\frac{4}{25}$	Square the numerator and square the denominator.
29	MA.6.NSO.3.5 challenging	$\frac{4}{5}$	A percent means hundredths, so write it over 100 and simplify.
30	MA.6.NSO.3.4 challenging	396	Multiply all of the prime factors together.
31	MA.6.NSO.1.4 challenging	7	Take each absolute value first, then subtract.
32	MA.6.GR.1.3 moderate	(-4, 8)	The fourth vertex lines up with A vertically and with C horizontally.
33	MA.6.NSO.1.3 challenging	$4 \frac{3}{4}$	The distance from 0 is the absolute value of the number.
34	MA.6.NSO.1.2 easy	4.0, -2.0, 0.0	A gain is a positive integer, a loss is a negative integer, and 0 means no gain or loss.
35	MA.6.AR.3.4 challenging	40	Divide the part by the percent written as a decimal to find the whole.
36	MA.6.GR.2.2 challenging	56	Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.
37	MA.6.NSO.1.3 easy	3.0, -3.0	Two numbers are 3 units from 0: 3 and -3.
38	MA.6.DP.1.1 challenging	How old is my teacher?	A question with only one possible answer does not produce a data set.